

Experiment No.	3
Experiment Title	Loan Amount Prediction using Linear, Ridge, Lasso, and Elastic Net Regression ¹
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1 Objective

To predict the loan sanction amount for applicants using Linear Regression and its regularized variants (Ridge, Lasso, Elastic Net), and to analyze the effect of L1/L2 regularization on model performance and coefficient behavior.

2 Problem Statement

Given applicant demographic, financial, and property-related attributes, the task is to predict a continuous target variable, the *Loan Sanction Amount (USD)*. This is a regression problem requiring numeric encoding of categorical variables, imputation of missing values, feature scaling, and comparison of baseline and regularized linear models using RMSE and R^2 .

3 Dataset Description

Dataset Name	Loan Amount Prediction
Dataset Source	Kaggle-style tabular dataset (<code>train.csv</code> / <code>test.csv</code>)
Number of Samples	30,000 (training set)
Number of Features	24 columns total (23 predictors + target), including numeric, categorical, and identifier fields
Number of Classes	N/A – continuous regression target (<i>Loan Sanction Amount (USD)</i>)
Missing Values	Present in several columns (e.g., Current Loan Expenses , Type of Employment , Dependents); rows with missing target were dropped, numeric NaNs imputed with median
Train-Test Split	80% train / 20% validation split from <code>train.csv</code> (<code>test.csv</code> has no target labels and was reserved for final prediction only)

4 Exploratory Data Analysis

Descriptive statistics revealed strong right-skew in several key numeric features. *Income (USD)* has a mean of 2568.32 but an extreme skewness of 167.25 and kurtosis of 28,618, indicating a small number of very high-income outliers dominating the distribution. *Loan Amount Request (USD)* is moderately right-skewed (skewness 1.26), while *Current Loan Expenses (USD)* contains negative values (minimum -999.0), suggesting placeholder/sentinel values used to encode missing or invalid entries.

- Dataset overview: 30,000 rows, 24 columns mixing numeric (age, income, loan amount, credit score, property price, etc.) and categorical (gender, profession, employment type, location) fields.
- Statistical summary: heavy right-skew in income and loan-related monetary fields; age is roughly symmetric (skewness 0.046).
- Missing value analysis: missing values present in employment type, dependents, and other columns, requiring imputation.
- Class distribution: N/A (regression task) – the target, *Loan Sanction Amount*, itself was examined via histogram/KDE instead.
- Correlation matrix: *Loan Amount Request*, *Credit Score*, and *Property Price* show the strongest positive linear relationships with the target, later confirmed by having the largest regression coefficients.
- Feature distribution: monetary features are right-skewed; categorical features (*Income Stability*, *Profession*, *Property Location*) show uneven category frequencies.

Inference: The presence of extreme skew and a sentinel value (-999) in *Current Loan Expenses* indicates that careful preprocessing (imputation and, ideally, outlier handling) is necessary before regression modeling. The strong visual relationship between *Loan Amount Request* and the target (seen in the scatterplot) foreshadows this feature becoming the dominant predictor in all four regression models, which is confirmed later in the coefficient comparison.

5 Data Preprocessing

- **Missing value handling:** Rows with a missing target value were dropped. Remaining numeric missing values were imputed using the column median (`fillna(X.median())`).
- **Encoding:** Categorical columns were one-hot encoded using `pd.get_dummies` with `drop_first=True` to avoid the dummy-variable trap. Identifier columns (*Customer ID*, *Name*) were dropped as non-predictive.
- **Feature scaling:** `StandardScaler` was fit on the training split and applied to both training and validation splits, standardizing all features to zero mean and unit variance.
- **Train-test split:** Since the provided `test.csv` has no target labels, an 80–20 train/validation split was carved out of `train.csv` using `random_state=42`.

- **Feature engineering:** No new derived features were created; all 23 predictor columns (after encoding) were used directly.

6 Machine Learning Algorithm

Linear Regression

Working principle: Models the target as a linear combination of input features: $\hat{y} = w^T x + b$, fit by minimizing the sum of squared residuals (Ordinary Least Squares). **Advantages:** Simple, interpretable, fast to train. **Limitations:** Sensitive to multicollinearity and outliers; can overfit when many correlated features are present.

Ridge Regression (L2)

Mathematical formulation: Minimizes $\sum (y_i - \hat{y}_i)^2 + \alpha \sum w_j^2$. Penalizes large coefficients without eliminating any feature, shrinking correlated coefficients together. **Advantages:** Reduces variance/overfitting, handles multicollinearity well. **Limitations:** Does not perform feature selection – all features are retained.

Lasso Regression (L1)

Mathematical formulation: Minimizes $\sum (y_i - \hat{y}_i)^2 + \alpha \sum |w_j|$. Drives some coefficients exactly to zero, performing implicit feature selection. **Advantages:** Produces sparser, more interpretable models. **Limitations:** Can be unstable when features are highly correlated (arbitrarily selects one of a correlated group); can underfit if α is too large.

Elastic Net

Mathematical formulation: Combines L1 and L2 penalties: $\sum (y_i - \hat{y}_i)^2 + \alpha (l1_ratio \sum |w_j| + (1 - l1_ratio) \sum w_j^2)$. **Advantages:** Balances Ridge's stability with Lasso's sparsity. **Limitations:** Introduces an additional hyperparameter (*l1_ratio*) to tune; convergence can be sensitive to feature scaling, as observed via `ConvergenceWarning` during fitting.

7 Experimental Setup

Python Version	3.14 (venv)
Scikit-Learn Version	Recent stable (<code>scikit-learn</code>)
IDE	Jupyter Notebook
Operating System	Linux
Hardware	Standard CPU (no GPU acceleration used)

8 Hyperparameter Selection

Parameter	Value
Ridge α search space	0.01, 0.1, 1, 10, 100
Lasso α search space	0.01, 0.1, 1, 10, 100
Elastic Net α search space	0.01, 0.1, 1, 10
Elastic Net $l1_ratio$ search space	0.2, 0.5, 0.8
Best Ridge α	100
Best Lasso α	100
Best Elastic Net	$\alpha = 0.1$, $l1_ratio = 0.5$

9 Results

Metric	Value (Best Model: Lasso, $\alpha = 100$)
RMSE (test/validation)	31,813.40
R^2 (test/validation)	0.5541
MAE (test/validation)	21,527.71
ROC-AUC	N/A (regression task)

Table 2: Validation-set performance comparison across all four models

Model	RMSE	R^2
Linear Regression	31,846.38	0.5532
Ridge Regression	31,829.16	0.5537
Lasso Regression	31,813.40	0.5541
Elastic Net	31,861.27	0.5528

Table 3: 5-Fold cross-validation performance (training set)

Model	MAE	RMSE	R^2
Linear Regression	21,496.61	31,661.09	0.5707
Ridge Regression	21,515.63	31,450.50	0.5766
Lasso Regression	21,451.47	30,860.63	0.5920
Elastic Net	21,696.42	31,117.83	0.5853

10 Result Visualization

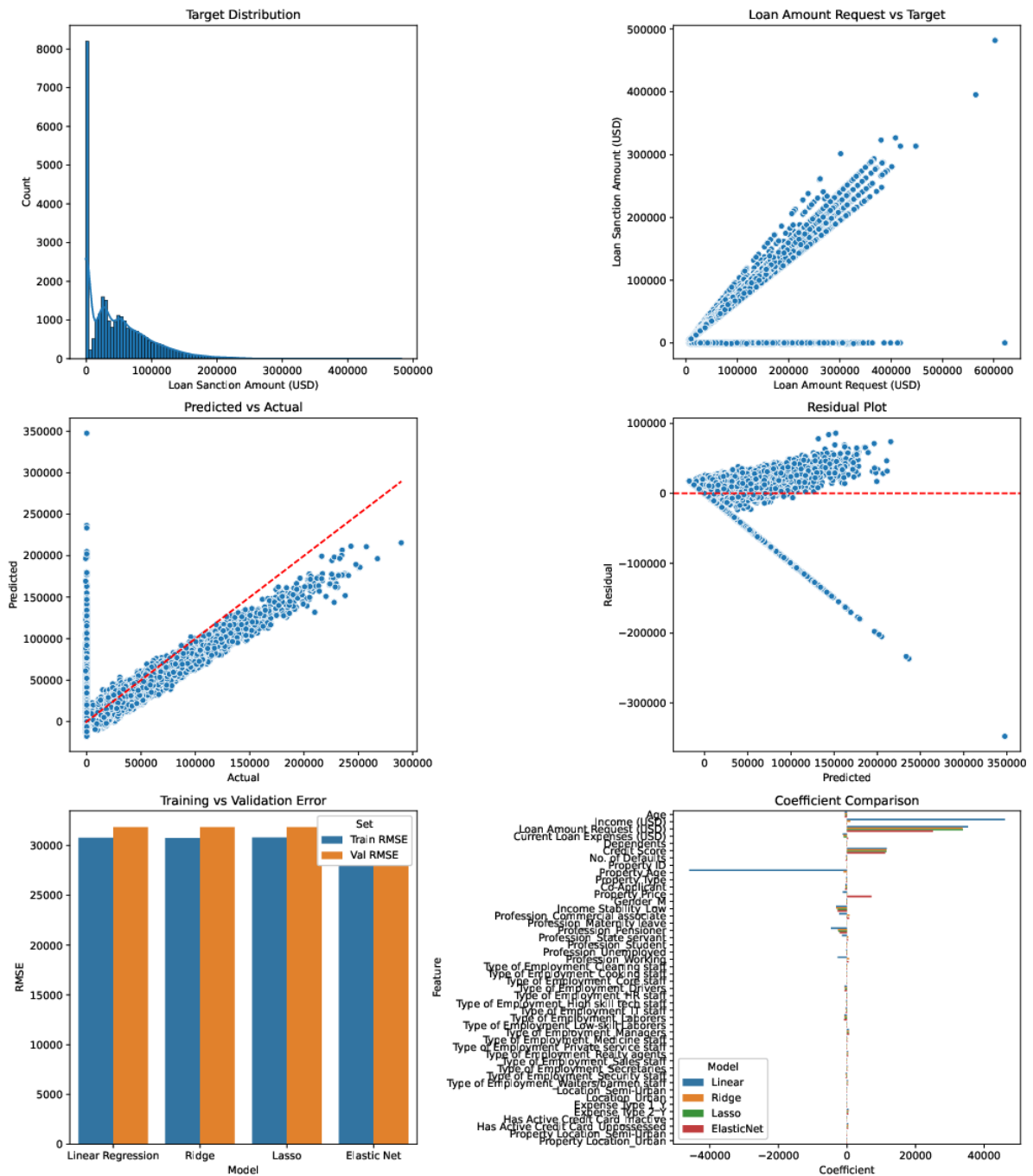


Figure 1: Target distribution, Loan Amount Request vs. Target scatterplot, Predicted vs. Actual (Ridge), residual plot, training vs. validation RMSE by model, and coefficient comparison across all four regression models.

Inference: The target distribution (top-left) is right-skewed, consistent with the descriptive statistics observed during EDA, suggesting a log-transform of the target could further improve model fit in future work. The scatterplot of *Loan Amount Request* vs. the target (top-right) shows a clear positive linear trend, explaining why this feature dominates the regression coefficients. The Predicted-vs-Actual plot for Ridge shows points reasonably

clustered around the diagonal for moderate loan amounts but increasing spread at higher values, indicating the model under-predicts for very large loans. The residual plot shows a fan-shaped (heteroscedastic) pattern – residual variance increases with predicted value – which violates the constant-variance assumption of OLS and explains the moderate R^2 (~ 0.55) across all models. The training-vs-validation RMSE bar chart shows minimal gap between training and validation error for all four models, indicating no severe overfitting. The coefficient comparison confirms that *Loan Amount Request*, *Credit Score*, and *Property Price* carry by far the largest (positive) coefficients across all models, while Lasso zeroes out several weaker features (e.g., *Dependents*, *Property Age*, *Property Price* in the Lasso column), demonstrating its feature-selection behavior.

11 Discussion

All four models achieved very similar validation RMSE (within ~ 50 units of one another, out of a target scale in the tens of thousands), indicating that regularization provided only a marginal improvement over plain Linear Regression on this dataset – Lasso ($\alpha = 100$) was the best performer, narrowly beating Ridge and Linear Regression. This suggests the feature set does not suffer from severe multicollinearity or overfitting at this sample size (24,000 training rows), limiting the benefit regularization can provide. However, cross-validation R^2 values (0.57–0.59) were consistently higher than validation R^2 values (0.55), which is expected since cross-validation averages over multiple folds and is a more robust generalization estimate.

The coefficient comparison highlights an important qualitative difference: Lasso’s ability to zero out weak predictors (*Dependents*, *Property Age*, *Property Price* partially) makes it more interpretable than Ridge, which shrinks but retains all coefficients. The convergence warnings observed for Elastic Net and Lasso during grid search suggest that further tuning of `max_iter` or additional feature scaling refinement could improve numerical stability, though it did not prevent the models from reaching competitive solutions.

The moderate R^2 (~ 0.55 – 0.59) across all models indicates that roughly 41–45% of the variance in loan sanction amount remains unexplained by this linear feature set, pointing to either non-linear relationships (better captured by tree-based models) or unobserved factors (e.g., lender-specific policies) influencing the sanctioned amount.

12 Conclusion

Linear Regression and its three regularized variants (Ridge, Lasso, Elastic Net) were implemented to predict loan sanction amounts. Lasso Regression ($\alpha = 100$) achieved the best validation performance (RMSE 31,813.40, R^2 0.5541), narrowly outperforming Ridge and plain Linear Regression, while also providing implicit feature selection by zeroing out several weak predictors. The experiment demonstrates that *Loan Amount Request*, *Credit Score*, and *Property Price* are the dominant drivers of the sanctioned loan amount, and that regularization offers modest but consistent gains in generalization on this dataset.

13 References

- [1] F. Pedregosa *et al.*, “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [2] Scikit-learn Developers, “Linear Models,” https://scikit-learn.org/stable/modules/linear_model.html.
- [3] Scikit-learn Developers, “Model Selection: GridSearchCV,” https://scikit-learn.org/stable/modules/grid_search.html.